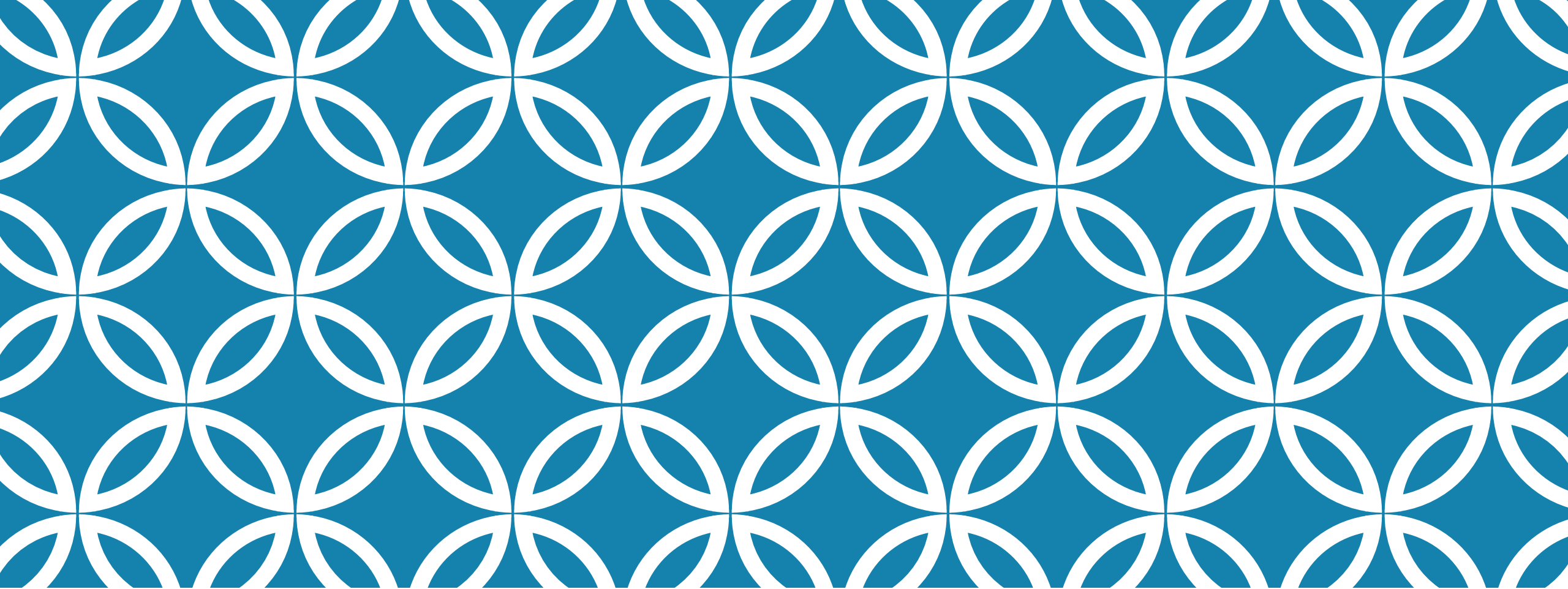




DETAILED ESTIMATE

Lecture II

HOW TO PREPARE A DETAILED ESTIMATE

- The unit quantity method is followed to prepare a detailed estimate.
- In this method the rates per unit work of one item (e.g. 1:3 concrete) including profit are considered first and the total cost for the item is found by multiplying the cost per unit of the rate by the number of units.
- The procedure for the preparation of a detailed estimate is divided into 2 PARTS.

a) Details of measurement & calculation of quantities - Dimensions of all individual items are taken from drawings & entered in the columns of a STANDARD MEASUREMENT FORM

Item no.	Description or particulars	No.	Length	Breadth	Height	Content or quantity

HOW TO PREPARE A DETAILED ESTIMATE

b) Abstract of Estimated Cost -

- The cost of each and every individual item is calculated by multiplying the quantity computed in the measurement form with the specific rate in a tabular form known as ABSTRACT FORM.
- The estimated cost is increased by 3% to 5% for any unforeseen expenditure and is called CONTINGENCY FUND.
- To maintain additional supervising staff at the work site and is called WORK CHARGED ESTIBALISHMENT a further amount 2.5% is added.
- For big projects an amount of 1% to 1.5% of the estimated cost is also provided to purchase special tools & plants for specific purpose.

Item no.	Description or particulars	Quantity	Unit	Rate Rs p	Unit of rate	Amount Rs p

CONTENTS OF DETAILED ESTIMATE

The detailed estimate is accompanied by :-

- Report
- Specifications
- Drawings (plan, section, elevation, site plan/key plan)
- Design charts and calculations
- Particulars of rates

FUNCTIONS OF AN ABSTRACT OF ESTIMATE

- Total estimated cost & different items can be known.
- This is the basis on which % rate tenders are called after excluding the amounts for contingency & work-charged establishment.
- It is part of a tender document- a contractor can arrive at his own rates from the schedule of work described in the description column.
- This is the basis on which bills are prepared for payment.
- Comparative costs of different items of work can be known.

CONTINGENCIES

- Contingencies indicate the expenses of a miscellaneous character which cannot be reasonably predicted during preparation of the estimate
- And to meet such unforeseen expenses an additional amount of 3% to 5% (3% - CPWD, 5% - PWD) of the estimated cost of works is provided in the total estimate.

COMPLETE SET OF ESTIMATE

- Title page indicating the name of the Engg. Dept, division, names of works & total estimated amount.
- Index of contents.
- Report
- Design data & calculations.
- Specifications
- Detailed statement of measurement , quantities & rates.
- Analysis of rates for unscheduled item.
- The detailed estimate in specified form.
- An abstract of estimate showing the total estimated cost for all items, contingencies, work-charged establishment, tools & plants, departmental charges.
- General abstract (in case of several units of work under a project)
- Drawings consisting of plans, detailed drawings, site plan, index plan etc.